

Solution Set 4 (supplemental)

- ① $t_{ij} \quad \left. \begin{matrix} i=1,2,3 \\ j=1,2,3 \end{matrix} \right\}$ form 9-components of a Cartesian tensor.

Remembering that t_{ij} transforms like $x_i x_j$, we can easily construct spherical tensors.

$$t_{11} + t_{22} + t_{33} \text{ is like } x_1^2 + x_2^2 + x_3^2 \text{ which is a scalar.}$$

$$\text{So } T_0^0 = t_{11} + t_{22} + t_{33}.$$

The three components

$$V_3 = t_{12} - t_{21}, \quad V_2 = t_{31} - t_{13}, \quad V_1 = t_{23} - t_{32}$$

transform like a Cartesian vector. Thus

$$T_1^0 = t_{12} - t_{21}$$

$$T_1^1 = -\frac{1}{\sqrt{2}} (t_{23} - t_{32} + it_{31} - it_{13})$$

$$T_1^{-1} = +\frac{1}{\sqrt{2}} (t_{23} - t_{32} - it_{31} + it_{13})$$

Now Y_2^m transform like spherical tensor of rank 2.

$$Y_2^{\pm 2} = \left(\frac{15}{32\pi}\right)^{1/2} \sin^2 \theta e^{\pm i2\phi} = \left(\frac{15}{32\pi}\right)^{1/2} \frac{(x \pm iy)^2}{r^2}$$

$$Y_2^{\pm 1} = \mp \left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{\pm i\phi} = -\left(\frac{15}{8\pi}\right)^{1/2} \frac{z(x \pm iy)}{r^2}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2} (3\cos^2 \theta - 1) = \left(\frac{5}{16\pi}\right)^{1/2} \frac{(3z^2 - r^2)}{r^2}$$

This means

$$T_2^{\pm 2} = \sqrt{\frac{3}{4}} (t_{11} - t_{22} \pm i(t_{12} + t_{21}))$$

$$T_2^{\pm 1} = -\frac{\sqrt{3}}{2} (t_{31} \pm i t_{32} + t_{13} \pm i t_{23})$$

$$T_2^0 = \sqrt{\frac{1}{2}} (2t_{33} - t_{11} - t_{22})$$

Problem 2: We can't construct $D_{mm}^1(R)$ explicitly.

First let us note that-

$$D^1(R) = \exp\left(-\frac{i}{\hbar} \Theta J_x^{(1)}\right)$$

where $J_x^{(1)}$ is the angular momentum operator for spin-1.

$$J_+^{(1)} |1m\rangle = \hbar \sqrt{(2+m)(1-m)} |1(m+1)\rangle$$

$$J_-^{(1)} |1m\rangle = \hbar \sqrt{(2-m)(1+m)} |1(m-1)\rangle$$

$$J_x^{(1)} = \frac{J_+^{(1)} + J_-^{(1)}}{2} \text{ which means explicitly.}$$

$$J_x^{(1)} \equiv \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \equiv |11\rangle$$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \equiv |10\rangle$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \equiv |1-1\rangle$$

$$(J_x^{(1)})^2 = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$(J_x^{(1)})^3 = \hbar^2 J_x^{(1)}$$

Hence we can show

$$D^{(1)}(R) = I + (\cos \theta - 1) \left(\frac{J_x^{(1)}}{\hbar^2} \right)^2 - i \sin \theta \left(\frac{J_x^{(1)}}{\hbar} \right)$$

Explicitly this means

$$D^{(1)}(R) = \begin{pmatrix} \frac{(\cos \theta + 1)}{2} & -\frac{i}{\sqrt{2}} \sin \theta & \frac{\cos \theta - 1}{2} \\ -\frac{i}{\sqrt{2}} \sin \theta & \cos \theta & -\frac{i}{\sqrt{2}} \sin \theta \\ \frac{\cos \theta - 1}{2} & \frac{i}{\sqrt{2}} \sin \theta & \frac{(\cos \theta + 1)}{2} \end{pmatrix}$$

12.5.4 (1) One can rotate $J_x^{(j)}$ and $J_y^{(j)}$ for that matter

$\hat{O} \cdot \vec{J}^{(j)}$ into $J_z^{(j)}$. i.e... the choice of $J_z^{(j)}$ is only

a convention any component of angular momentum can be chosen as the "z" component! Hence the eigenvalues will be the same.

(2) Any state in the space spanned by $|jm\rangle$ $m=-j \dots j$ can be expanded in this basis. This basis can be chosen to be eigenvector of J_z .

Thus

$$|\psi\rangle = \sum_{m=-j}^j \psi_j |jm\rangle$$

But then

$$(J-j\hbar)(J-(j-1)\hbar) \dots (J+j\hbar) |\psi\rangle = 0 \quad \text{as}$$

can be checked easily. Hence an operator acting on all states giving 0 is a "zero" operator.

(3) clearly

$$J^{2j+1} = \alpha_0 + \alpha_1 J + \alpha_2 J^2 + \dots + \alpha_{2j} J^{2j}$$

Hence

$$J^{2j+2} = \alpha J + \alpha_1 J^2 + \alpha_3 J^3 + \dots + \alpha_{2j} J^{2j+1}$$

↓

Can be
expanded
in $J^0 \dots J^{2j}$

$$\therefore J^{2j+2} = \alpha'_0 + \alpha'_1 J + \alpha'_2 J^2 + \dots + \alpha'_{2j} J^{2j}$$

and - similarly we can show any power of

J of the form J^{2j+k} can be written as a linear
combination of $J^0, J^1, \dots, J^{2j}$.

Exercise 15.3.3

(1) Using $\langle jj; j' | jj, 10 \rangle = \left(\frac{j}{j+1} \right)^{1/2}$ we have to

show

$$\langle \alpha j || J_1 || \alpha' j' \rangle = \delta_{\alpha\alpha'} \delta_{jj'} \hbar \sqrt{j(j+1)}$$

Now we know

$$\begin{aligned} J_1^0 &= J_z \\ J_1^+ &= -\frac{(J_x + iJ_y)}{\sqrt{2}} = -\frac{J_+}{\sqrt{2}} \\ J_1^- &= \frac{(J_x - iJ_y)}{\sqrt{2}} = \frac{J_-}{\sqrt{2}} \end{aligned}$$

Consider

$$\langle \alpha jj | J_z | \alpha' j' j' \rangle = \hbar \delta_{\alpha\alpha'} \delta_{jj'}$$

From Wigner-Eckart theorem

$$\langle \alpha jj | J_z | \alpha' j' j' \rangle = \langle \alpha jj | J_1^0 | \alpha' j' j' \rangle$$

$$= \langle \alpha j || J_1 || \alpha' j' \rangle \langle jj; j' | jj', 10 \rangle$$

∴ We see that

$$\langle \alpha j || J_1 || \alpha' j' \rangle \langle jj; j' | jj', 10 \rangle = \hbar \delta_{\alpha\alpha'} \delta_{jj'}$$

Setting $j=j'$ we see

$$\langle \alpha j || J_1 || \alpha' j' \rangle = \hbar \sqrt{j(j+1)} \delta_{\alpha\alpha'} \delta_{jj'}$$

(2) $\vec{J} \cdot \vec{A}$ is a scalar operator, which means

$$\langle \alpha' j m' | \vec{J} \cdot \vec{A} | \alpha j m \rangle = \langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle \underbrace{\langle j m'; j 0 | j m, 00 \rangle}_{\delta_{mm'}}$$

$\therefore$ The left hand side ^{is non-zero} only if $m = m'$. Further

$\langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle$ does not depend on m, m' . Hence we can set $m' = m = j$ to evaluate it. We see

$$\langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle = \langle \alpha' j j | \vec{J} \cdot \vec{A} | \alpha j j \rangle$$

$$= \langle \alpha' j j | J_z A_z + \frac{1}{2} (J_- A_+ + J_+ A_-) | \alpha j j \rangle$$

$$\begin{cases} \langle \alpha' j j | J_z = j \hbar \langle \alpha' j j | \\ \langle \alpha' j j | J_- = 0 \\ \langle \alpha' j j | J_+ = \hbar \sqrt{2j} \langle \alpha' j j-1 | \end{cases}$$

$$\begin{aligned} \langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle &= \hbar j \langle \alpha' j j | A_z | \alpha j j \rangle \\ &\quad + \hbar \sqrt{\frac{j}{2}} \langle \alpha' j j-1 | A_- | \alpha j j \rangle \end{aligned}$$

$$\begin{aligned} &= \hbar j \langle \alpha' j j | A_1^0 | \alpha j j \rangle \\ &\quad + \hbar \sqrt{\frac{j}{2}} \langle \alpha' j j-1 | \sqrt{2} A_1^{-1} | \alpha j j \rangle \end{aligned}$$

$$\langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle = \left[\hbar j \langle jj | jj, 10 \rangle + \hbar \sqrt{j} \langle jj-1 | jj-1-1 \rangle \right] \cdot \langle \alpha' j || A_1 || \alpha j \rangle$$

$\therefore$ substituting

$$\langle jj | jj, 10 \rangle = \sqrt{\frac{j}{j+1}} \quad (\text{given}).$$

$$\langle j(j-1) | jj-1-1 \rangle = \sqrt{\frac{1}{j+1}} \quad (\text{proved below}).$$

we get

$$\langle \alpha' j || \vec{J} \cdot \vec{A} || \alpha j \rangle = \hbar \sqrt{j(j+1)} \langle \alpha' j || A_1 || \alpha j \rangle$$

$$\therefore \langle \alpha' j m' || \vec{J} \cdot \vec{A} || \alpha j m \rangle = \hbar \sqrt{j(j+1)} \langle \alpha' j || A_1 || \alpha j \rangle \delta_{mm'}$$

Let us now show $\langle jj-1 | jj-1-1 \rangle = \sqrt{\frac{1}{j+1}}$.

Consider.

$$\langle jm | J_z^q | jm' \rangle = \langle j || J_1 || j \rangle \langle jm | jm', 1q \rangle.$$

Now when $m=j$ $m'=j$ $q=0$ we see L.H.S = $\hbar j$

$$\therefore \hbar j = \langle j || J_1 || j \rangle \cdot \underbrace{\langle jj | jj, 10 \rangle}_{\sqrt{\frac{j}{j+1}} \text{ (given)}} \quad \text{--- (1)}$$

when $m=j-1$ $m'=j$ $q=-1$ we get L.H.S = $\langle jj-1 | \underbrace{J_z^{-1}}_{J_z} | jj \rangle$

$$\Rightarrow \text{L.H.S} = \hbar \sqrt{j}$$

$$\hbar \sqrt{j} = \langle j || J_1 || j \rangle \langle jj-1 | jj, 1-1 \rangle \frac{1}{\sqrt{2}} \quad \text{--- (2)}$$

using eq. (1) and (2) we get the result.

(3) Now $\langle \alpha' j m' | A_1^q | \alpha j m \rangle = \langle \alpha' j || A_1 || \alpha j \rangle \cdot \langle j m' | j m, 1 q \rangle$

using the result from (2) we see R.H.S

$$= \frac{\langle \alpha' j m | \vec{J} \cdot \vec{A} | \alpha j m \rangle}{\hbar \sqrt{j(j+1)}} \cdot \langle j m' | j m, 1 q \rangle$$

Now

$$\langle j m' | J_1^q | j m \rangle = \underbrace{\langle j | J_1 | j \rangle}_{\hbar \sqrt{j(j+1)}} \langle j m' | j m, 1 q \rangle$$

R.H.S

$$= \frac{\langle \alpha' j m | \vec{J} \cdot \vec{A} | \alpha j m \rangle}{\hbar \sqrt{j(j+1)}} \cdot \frac{\langle j m' | J_1^q | j m \rangle}{\hbar \sqrt{j(j+1)}}$$

$$= \frac{\langle \alpha' j m | \vec{J} \cdot \vec{A} | \alpha j m \rangle}{\hbar^2 j(j+1)} \langle j m' | J_1^q | j m \rangle$$

15.3.4 We are given.

$$\vec{\mu} = \gamma_1 \vec{J}_1 + \gamma_2 \vec{J}_2$$

The state $|j, m; j_1, j_2\rangle$ is the total angular momentum eigenstates of $\vec{J} = \vec{J}_1 + \vec{J}_2$.

Using Eq. (15.3.19) in Shankar we get

$$\langle j, m; j_1, j_2 | \vec{J}_1 | j, m; j_1, j_2 \rangle$$

$$= \frac{\langle j, m; j_1, j_2 | \vec{J} \cdot \vec{J}_1 | j, m; j_1, j_2 \rangle}{\hbar^2 j(j+1)} \langle j, m; j_1, j_2 | \vec{J} | j, m; j_1, j_2 \rangle$$

$$\text{clearly } \langle J_x \rangle = \langle J_y \rangle = 0$$

$$\therefore \langle \mu_x \rangle = \langle \mu_y \rangle = 0$$

$$\langle \mu_z \rangle = \frac{\gamma_1}{\hbar^2 j(j+1)} \langle (\vec{J}_1)_z \rangle + \frac{\gamma_2}{\hbar^2 j(j+1)} \langle (\vec{J}_2)_z \rangle$$

$$= \frac{m \hbar}{\hbar^2 j(j+1)} \left\{ \gamma_1 \langle j, m; j_1, j_2 | \vec{J} \cdot \vec{J}_1 | j, m; j_1, j_2 \rangle + \gamma_2 \langle j, m; j_1, j_2 | \vec{J} \cdot \vec{J}_2 | j, m; j_1, j_2 \rangle \right\}$$

$$\langle j, m; j_1, j_2 | \underbrace{\vec{J} \cdot \vec{J}_1}_{\vec{J}_1^2 + \vec{J}_2 \cdot \vec{J}_1} | j, m; j_1, j_2 \rangle = \hbar^2 j_1(j_1+1) + \frac{\hbar^2}{2} (j(j+1) - j_1(j_1+1) - j_2(j_2+1))$$

$$\langle j, m; j_1, j_2 | \underbrace{\vec{J} \cdot \vec{J}_2}_{(\vec{J}_2^2 + \vec{J}_1 \cdot \vec{J}_2)} | j, m; j_1, j_2 \rangle = \hbar^2 j_2(j_2+1) + \frac{\hbar^2}{2} (j(j+1) - j_1(j_1+1) - j_2(j_2+1))$$

$$\langle \mu_z \rangle = m \hbar \left[\frac{\delta_1 + \delta_2}{2} + \frac{(\delta_1 - \delta_2)}{2} \frac{(j_1(j_1+1) - j_2(j_2+1))}{j(j+1)} \right]$$

(2) A proton in a $2P_{1/2}$ state is a proton with $\vec{L} = 1$ $\vec{S} = \frac{1}{2}$ $\vec{J} = \frac{1}{2}$.

Thus.

$$\vec{\mu}_{\text{proton}} = \frac{ge}{2Mc} \vec{S} + \frac{e}{2Mc} \vec{L} \quad (g=5.6)$$

using the formula obtained earlier

$$\langle \mu_z \rangle = \pm \frac{\hbar}{2} \cdot \frac{e}{2Mc} \left[\frac{(g+1)}{2} + \frac{(g-1)}{2} \frac{(\frac{3}{4} - 2)}{\frac{3}{4}} \right]$$

$$= \pm \frac{1}{4} \left[(g+1) - \frac{5}{3} (g-1) \right] \frac{\hbar e}{2Mc}$$

Nuclear
Magneton

Substituting $g=5.6$ we get

$$|\langle \mu_z \rangle| \sim \left| \frac{1}{2} \left[\frac{4}{3} - \frac{g}{3} \right] \right| \text{ Nuclear Magnetons}$$

$$\approx 0.26 \text{ Nuclear Magnetons}$$

(3) For an electron in $2P_{1/2}$ state we can use the same procedure as above and we get

$$\langle \mu_z \rangle \sim \frac{1}{2} \left[\frac{4-g}{3} \right] \text{ Bohr Magnetons} \quad \text{where } g=2$$

$$\approx \frac{1}{3} \text{ Bohr magnetons}$$